

Assignment 01 & 02

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Ans to the ques no 01

a.

Node	h(n)	parent
S	12	ϕ
C	10	S
A	14	S
F	9	S
D	8	F
E	15	F
H	17	D
B	16	C
G	0	A

Priority Queue	Closed
S	S
A	F
E	D
F	C
D	A
E	G
H	
B	
G	

Shortest Route: $S \rightarrow A \rightarrow G$ || Cost = 15

b.

Node	g(n)	h(n)	f(n)	parent
S	0	12	12	ϕ
A	7	14	21	S
C	5	10	15	S
F	2	9	11	S
D	5	8	13	F
E	4	15	19	F
H	10	17	27	D E
B	14	16	30	C
G	15	0	15	A

Priority Queue	Closed
S	S
A	F
E	D
F	C
D	E
E	A
H	G
B	
G	

Shortest Route: $S \rightarrow A \rightarrow G$ || Cost = 15

The GBFS route and A* search routes are both the same with similar costs. Reasons: Some heuristic value of nodes are not admissible as $h(n) \not\leq h^*(n)$, creating a non optimal path

Ans to the ques no-02

Eucledian Distance will expand fewer nodes

In Eucledian Distance,

$$h_2(n) = \sqrt{(4x)^2 + (4y)^2} \text{ ——— } \textcircled{1}$$

eq ① always a lower bound as the problem allows diagonal move making it admissible and consistent.

But in Manhattan distance,

$$h_1(n) = 4x + 4y \text{ ——— } \textcircled{11}$$

Eq ① assumes orthogonal moves only. Because diagonal moves are allowed and cheaper than two orthogonal moves, $h_1(n)$ estimates the true cost. Thus, h_1 is not admissible in this problem.

Ans to the ques no 03

Node	$h(n)$
A	8
B	4
C	4
D	1
E	0

Starting from (top-left)

1	4	3	2
6	10	5	1
2	9	8	4
3	1	8	15
5	7	3	6
4	6	5	10

↓
stuck/stops
at
10
↓
local
optima.

starting from (top-right)

1	4	3	2
6	10	5	1
2	9	8	4
3	1	8	15
5	7	3	6
4	6	5	10

↓
stops at
10
↓
local
optima

starting from (bottom right)

1	4	3	2
6	10	5	1
2	9	8	4
3	1	8	15
5	7	3	6
4	6	5	10

↓
stops at
10
↓
local
optima

Starting from (bottom-left)

1	4	3	2
6	10	5	1
2	9	8	4
3	1	8	15
5	7	3	6
4	6	5	10

↓
stops at
15
↓
global
optima

Hill climbing does not guarantee the best answer.

(1) Looks at neighbouring cells only

(2) Moves greedily to a better neighbour, never accept worse.

(3) It stops when no neighbour is better → that point is a local maximum, which might not be the global maximum.

Ans to the quesno -05

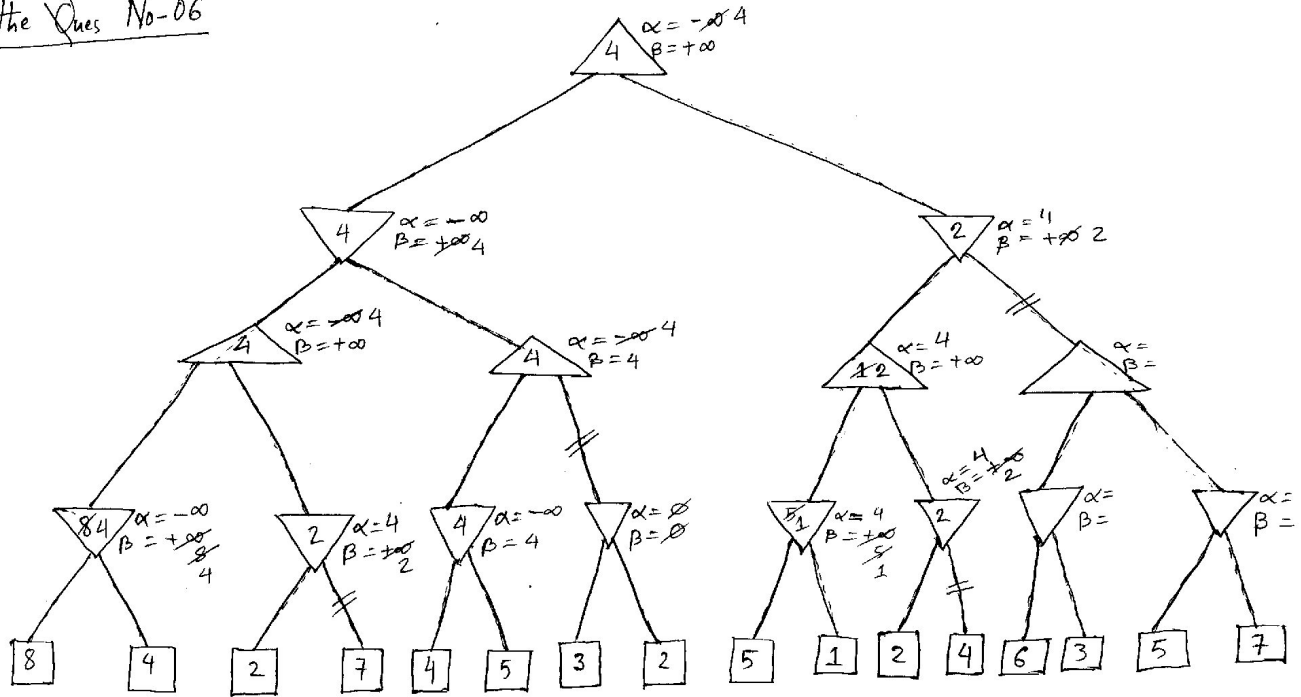
According to Boltzmann's Distribution,

$$p = e^{\frac{-\Delta E}{T}}$$

When 'temperature', T is high/large, the probability of 'bad' moves are more likely to be allowed at the start.

In time, when temperature decreases, the probability of choosing 'bad' moves decreases.

Ans to the Ques No-06



Ans to the ques no-07

Encoding Scheme

- (1) Length = number of stations + 1
- (2) the starting & ending elements are same
- (3) No repeated elements in intermediate elements (-between start and end)
- (4) The starting/ ending station cannot appear in intermediate elements

Random generated 6 populations:

$$\text{fitness } f^n = \frac{1}{1 + \sum (\text{distance})}$$

$$A C E B H D F G A \longrightarrow \frac{1}{38+1} = 0.0257$$

$$A D F G B C H E A \longrightarrow \frac{1}{37+1} = 0.0253$$

$$A H B D F C G E A \longrightarrow \frac{1}{48+1} = 0.0204$$

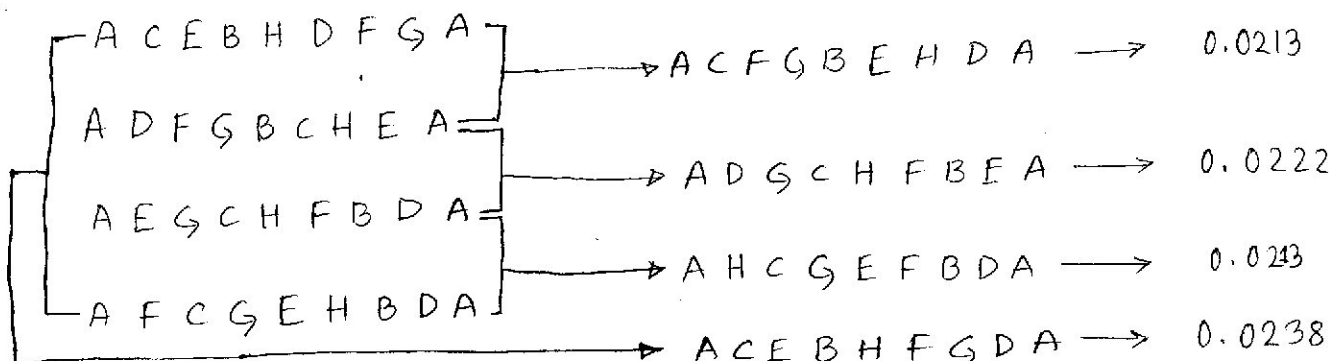
$$A E G C H F B D A \longrightarrow \frac{1}{4+1} = 0.0212$$

$$A B D H F G C E A \longrightarrow \frac{1}{45+1} = 0.0217$$

$$A F C G E H B D A \longrightarrow \frac{1}{42+1} = 0.0232$$

Children Generation

fitness



Mutated Children

Fitness

A C F G B E H D A $\rightarrow$ A E F G B C H D A $\rightarrow$ 0.0238

A D G C H F B E A $\rightarrow$ A D C G H F B F A $\rightarrow$ 0.02

A H C G E F B D A $\rightarrow$ A H D G E F B C A $\rightarrow$ 0.0213

A C E B H F G D A $\rightarrow$ A C E B G F H D A $\rightarrow$ 0.0213

Analyzing parents, children & mutated children

A D F G B C H E A having 0.0263 fitness value is the best solution after one iteration.

Population for next iteration using 2-elitism

A D F G B C H E A
A C E B H D F G A } Best 2

A E F G B C H D A
A D C G H F B E A } Random 2.

Ans to the Ques no - 08

$$f(x) = x^4 - 14x^2 + 24x + 30$$

We will use binary encoding scheme.

Random generated population (4):

0	: 0000
3	: 0011
5	: 0101
9	: 1001

fitness function, $z(x) = \frac{1}{1+f(x)}$

$f(0) = 30$	$z(0) = \frac{1}{31} = 0.0322$
$f(3) = 57$	$z(3) = 0.0175$
$f(5) = 425$	$z(5) = 0.00235$
$f(9) = 5673$	$z(9) = 0.000176$

# Random chosen parents :	# Random crossover	# Mutation
0000 ≈ 0 0011 ≈ 3	$\begin{array}{c c} 0000 & 0001 \approx 1 \\ 0011 & 0010 \approx 2 \end{array}$	0001 $\rightarrow$ 0110 ≈ 6 0010 $\rightarrow$ 1010 ≈ 10

$f(1) = 41$	$z(1) = 0.0238$	# Analyzing parents, children & mutated children $0 \approx 0000$ is the best solution # population in next generation with 2-elitism $0000 \sim p_1$ $0010 \sim c_2$
$f(2) = 38$	$z(2) = 0.0256$	
$f(6) = 466$	$z(6) = 0.00103$	
$f(10) = 8870$	$z(10) = 0.0001127$	

Ans to the ques no-09

a. Encoding scheme: we have 8 seats around a circular table. A chromosome is a permutation of the 8 guests. As it is circular, first and last positions are neighbours.

4 parent chromosome:

$$C_1 = 1, 2, 4, 5, 6, 3, 7, 8$$

$$C_2 = 1, 4, 2, 5, 3, 6, 7, 8$$

$$C_3 = 1, 5, 4, 2, 3, 6, 8, 7$$

$$C_4 = 1, 3, 6, 5, 4, 2, 7, 8$$

b. fitness function:

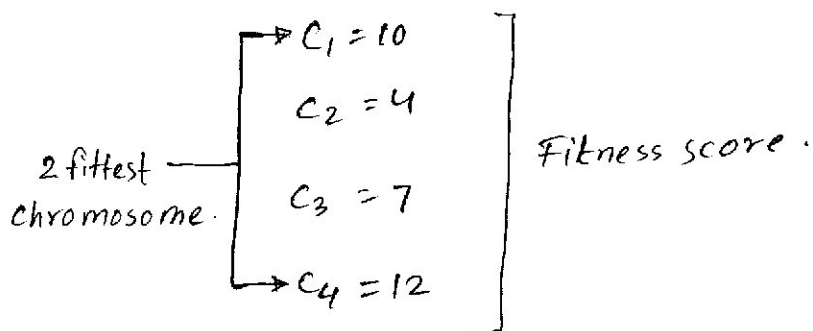
+1 for each neighbour that is in their "like" set.

-1 " " " " " " "avoid" "

0 otherwise

So, for each guest, maximum score is +2
minimum score is -2

$$\therefore \text{Fitness} = \sum_{i=1}^8 \text{score}_i \quad | \text{ fitness} \propto \text{better seating.}$$



c. No, standard chromosome will usually break the permutation constraint. There is a possibility of duplicates and missing guests, violating the constraints.

Rather we will use ordered crossover.

1. **True** :

2. False : Simulated annealing does not guarantee better performance than hill climbing in all search spaces. Its advantage depends on the problem's landscape. In some cases, the stochastic exploration helps escape local optima, but in smooth or simple landscapes, it might perform no better, even worse, due to random moves.
3. False : When the initial temperature is very high, simulated annealing behaves almost like random search, not hill climbing. Hill climbing only accepts better moves, while high temperature simulated annealing accepts almost any move, giving it much higher randomness.
4. False : Random restarts with hill climbing can increase the chance of finding the global optimum, especially in landscape with many local maxima. Each restart begins the search in a new region, improving the probability of eventually hitting the global peak.

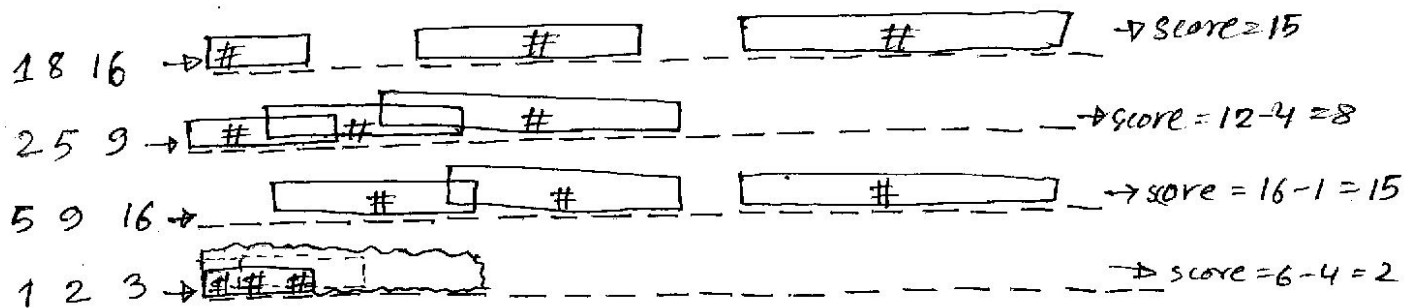
e. False: A hill climbing algorithm that stops after a fixed number of iterations can find the global optimum if it happens to reach it before stopping. Increasing the number of iterations increase the chance of finding it; it doesn't make it impossible.

Ans to the ques no-011

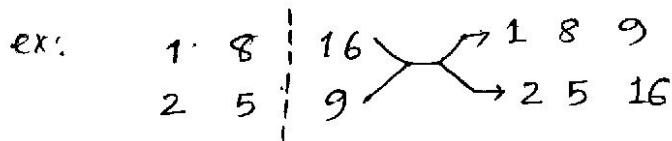
a. The chromosome will consist of 3 elements (length=3) and each element will represent the position of the camera.

b. Fitness function = total coverage - total clash

c. Randomly Generated Population (with fitness)



d. Single Point Crossover will work.



Ans to the ques no-12

a. Encoding Scheme: Use a binary chromosome of length d , one gene per candidate.

example: $[0, 0, 0, 1, 1, 1]$

b. fitness function:

→ Reward similarity between different subset = x

→ Penalize " " Same " = y

$$\therefore \text{Fitness} = x - y$$

ex: $[0, 0, 0, 1, 1, i]$

$$\therefore S_0 = \{A_1, A_2, A_3\}, S_1 = \{A_4, A_5, A_6\}$$

$$\therefore x = 6 + 5 + 7 + 4 + 2 + 3 + 9 + 6 + 2 = 44$$

$$\therefore y_{s_0} = 8 + 3 + 7 = 18 \quad | \quad \therefore y_{s_1} = 7 + 1 + 5 = 13$$

$$\therefore y = 18 + 13 = 31$$

$$\therefore \text{fitness} = 44 - 31 = 13$$

c. let, $P_1 = 000/111$
 $P_2 = 10/001$ } using single point crossover

$$\therefore C_1 = 000001$$

$$\therefore C_2 = 101111$$

Ans

a. As we have 5 items, we will use 5 bit binary string. Any state whose total volume > 1 is invalid and must be rejected.

When we choose no food item at all, the start state will be 00000 (all zeros).

b. From a current state, we generate neighbouring elements by adding exactly one more item provided the new state is still within the 1 liter limit.

If we ever reach a state with total volume = 1L and no additional item can be added without exceeding 1L, then that state has no neighbors and local search stops.

c. Hill climbing

→ looks for neighbour

→ if no better neighbour exists with higher nutritional value, it stops. Else it moves to neighbour

Let,
current state = 10100

∴ value = Sandwich + Yogurt
= 60 + 50 = 110

neighbours: 10000 → value = 60
00100 → value = 50 } worse

∴ Stuck at 10100.

Simulated Annealing

→ Prefers better neighbour

→ Initially allows worse moves, matures over time.

→ Uses $p = e^{-4E/T}$

At $T = 100$,

current = 10100 → value = 110

neighbour's value = 60

∴ $\Delta = 60 - 110 = -50$

∴ $p = e^{-|\Delta E|/T}$

= $e^{-50/100} = e^{-0.5} = 0.6065 = 60.6\%$

∴ 60.65% possibility it will accept worse move, shifting to neighbour's worse value.

Ans to the ques no-14

a. $h_5(n) = 2 \times h_2(n)$ & $h_9(n) = \max(h_2(n), h_3(n))$

b. h_7 is the dominating one

c. h_7 is the best heuristic.

Ans to the ques no-15

